

The empirical recurrence for OEIS A259721, recovered from its tail

Adrian Perez Fontelles
Independent researcher

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Abstract

OEIS A259721 records “Empirical recurrence of order 84” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 106$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 2$, so the recurrence is proved for every $n \geq 86$. Its last coefficient is $c_{84} = 1 \neq 0$, so no further tail satisfies anything shorter; any order-84 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A259721 is “Number of $(n+2) \times (6+2)$ 0..1 arrays with each 3×3 subblock having clockwise perimeter pattern 00000001 or 00000011.”. It has offset 1 and begins

722, 1219, 2858, 5458, 7076, 16294, 33846, 51254, ...

The entry states:

Empirical recurrence of order 84 (see link above)

As of the “Last modified” line on the live entry (Jun 02 2025 11:54:25, revision 6) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 106$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 106$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 84: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 2$ is the first whose minimal order is exactly the stated 84.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 212$ exact terms from $n = 2$ on, it returns order 84 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{84} c_i a(n-i),$$

c_1	0	c_{22}	−70722	c_{43}	−341342	c_{64}	2800
c_2	1	c_{23}	−6563	c_{44}	−494802	c_{65}	1129
c_3	12	c_{24}	103350	c_{45}	−396861	c_{66}	5040
c_4	19	c_{25}	180821	c_{46}	−74964	c_{67}	−19675
c_5	−11	c_{26}	130029	c_{47}	354975	c_{68}	1925
c_6	−78	c_{27}	−38252	c_{48}	379389	c_{69}	346
c_7	−197	c_{28}	−242740	c_{49}	245721	c_{70}	−2385
c_8	−120	c_{29}	−328950	c_{50}	14775	c_{71}	4817
c_9	341	c_{30}	−169963	c_{51}	−317553	c_{72}	−1027
c_{10}	1031	c_{31}	133056	c_{52}	−225614	c_{73}	−330
c_{11}	1404	c_{32}	407167	c_{53}	−107720	c_{74}	582
c_{12}	−43	c_{33}	453797	c_{54}	11191	c_{75}	−777
c_{13}	−3531	c_{34}	168858	c_{55}	228952	c_{76}	231
c_{14}	−6670	c_{35}	−237805	c_{56}	99270	c_{77}	84
c_{15}	−5532	c_{36}	−519274	c_{57}	34323	c_{78}	−75
c_{16}	4555	c_{37}	−506381	c_{58}	−5936	c_{79}	70
c_{17}	19864	c_{38}	−149557	c_{59}	−131390	c_{80}	−22
c_{18}	26608	c_{39}	305371	c_{60}	−29364	c_{81}	−8
c_{19}	11811	c_{40}	543643	c_{61}	−8153	c_{82}	3
c_{20}	−29173	c_{41}	487481	c_{62}	−3467	c_{83}	−2
c_{21}	−72024	c_{42}	125632	c_{63}	58606	c_{84}	1

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 86$, and it is the recurrence OEIS A259721 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 2$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{84} - c_1x^{84-1} - \dots - c_{84}$. Its constant term is $-c_{84} = -1$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k\lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 84 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 84, so $\deg q = 84$, and it is monic. It holds from some index t on. From $\max(t, 2)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 84, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 26 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 84, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 1.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-84 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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